

Disambiguating “Aspect” and “Cell” Across the Trilogy

Series T — Trilogy Ambiguity Reduction | Note T3.02 | #700

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

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Attribution

This work derives from and formalizes commitments across the CKS theory trilogy: *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026), and *Inter-Self Coordination via Shared Substrate / Full Aspect Integration* (Li, April 2026). It introduces no new axioms. Its contribution is to articulate, in operational form, the precise scope-specific meanings of the terms *cell* and *aspect* as those terms are used across all three source papers, and to establish the architectural commitment that connects those meanings — so that practitioners reading the trilogy can use the terms without ambiguity and so that the prior-art record is explicit about what these terms name.

Abstract

The CKS trilogy uses the term *cell* in all three papers and the term *aspect* in Papers 2 and 3. Both terms are load-bearing. Each carries a precise meaning that is consistent across papers, but the meanings are easy to misread across the trilogy’s scope shifts: a reader encountering *cell* in Paper 3 might not recognize it as the same architectural object Paper 1 defined, and a reader encountering *aspect* in Paper 3 might miss that the exchange unit role is additive rather than definitional replacement. This note establishes the scope-specific meaning of each term in each paper, the architectural commitment that unifies the meanings, and the Paper 3 context-specific term *contributed aspect*, which names the specific condition of an aspect that is actively participating in a shared substrate event. The prior-art position: *cell* is defined in Paper 1 and that definition holds unchanged through Papers 2 and 3; *aspect* is defined in Paper 2 and Paper 3 adds an exchange unit role to the existing definition without replacing it; *contributed aspect* is a Paper 3 context-specific qualifier, not a new architectural tier.

1. Concept Identification

Terms under disambiguation: *cell*, *aspect*, *contributed aspect*

Status in the trilogy:

- *Cell* — defined in Paper 1 as the atomic governance unit; extended in Paper 2 with explicit articulation of internal structure and placement in the three-tier hierarchy; unchanged in Paper 3, where cells participate in inter-Self coordination indirectly through their parent aspects.
- *Aspect* — first defined in Paper 2 as a coordination unit grouping cells within a Self; extended in Paper 3 with the exchange unit role in Full Aspect Integration events.

- *Contributed aspect* — a Paper 3 context-specific term naming an aspect that has been contributed to a shared substrate during a Full Aspect Integration (FAI) event; not a separate architectural tier.

These three terms are treated together in this note because they define the structural tiers of the CKS architecture and because their disambiguation is closely coupled: the cell is the base unit, the aspect groups cells, and the contributed aspect is the form an aspect takes when it crosses the inter-Self perimeter.

2. Cell Across the Trilogy

Paper 1 definition.

The cell is the atomic governance unit of the CKS architecture. Its defining properties at Paper 1 scope: one LLM instance operating as the instinct layer; one harness substrate configuring instinct layer behavior; one governance substrate with a DNA layer (orchestration rules governing cell behavior) and an action layer (operational records of the cell's execution history); governed by one governance authority holding the three rights (inspect, modify, override) over all substrate content and orchestration rules. The cell is the primary named structural unit in Paper 1. "Aspect" and "Self" do not appear at Paper 1 scope. The cell's modular, composable design is explicit in Paper 1, and the six architectural commitments (substrate-based, human-governed, conflict-preserving, AI-as-mediator, tool-agnostic, linear-cost) all hold at cell scope.

Paper 2 definition.

The cell is the same architectural object — the atomic governance unit, now explicitly placed at the base of the three-tier hierarchy (cells → aspects → Self). Paper 2 articulates structure that was implicit in Paper 1: the DNA layer and action layer are named and differentiated; lifecycle (birth, mating, death) is specified; content-domain specification and relational-role determination are added. The cell object is unchanged. Paper 2 does not redefine the cell; it makes explicit what Paper 1's modular unit required. All six Paper 1 commitments hold at cell scope without modification. The addition of tiers above the cell (aspects and Self) does not alter what a cell is; it places the cell in a larger architecture.

Paper 3 definition.

The cell is the same architectural object as in Papers 1 and 2 — the atomic governance unit at the base of the governance architecture. At Paper 3 scope, cells continue to operate within each Self's home perimeter exactly as Paper 2 specifies. When a Self participates in an FAI event, it contributes aspects to the shared substrate; the aspect surfaces its constituent cells' DNA-layer and action-layer content. Cells participate in FAI indirectly — through their parent aspect — not as direct contributors. The cell's internal structure, its governance properties, and its base definition are unchanged. Paper 3 adds no new properties to the cell itself; the only Paper 3 novelty concerning cells is the indirect participation path.

The cell's architectural consistency.

The cell is the most stable concept in the trilogy. Its fundamental definition is set in Paper 1 and never modified. Paper 2 enriches the description of what a cell contains (DNA layer, action layer, lifecycle,

content-domain) without changing what the cell is. Paper 3 adds context (FAI participation through parent aspects) without changing what the cell is. This consistency is a strong prior-art position: the cell governance architecture was established in Paper 1, and Papers 2 and 3 build on it without redefinition. Any implementation treating the Paper 2 or Paper 3 cell as a different architectural object from the Paper 1 cell is misreading the trilogy.

3. Aspect Across the Trilogy

Paper 1 scope.

“Aspect” does not appear in Paper 1. There is no prior usage of the term in the trilogy before Paper 2. This is a deliberate scope boundary: Paper 1 establishes the cell as the sole named structural unit. Aspects and Selves are Paper 2 concepts introduced when cells are composed into larger structures.

Paper 2 definition.

An aspect is a coordination arrangement of cells organized around a shared content-domain. Its defining properties: a DNA layer containing aspect-level coordination rules that govern the cells within the aspect; an action layer recording the aspect’s operational history; a content-domain specification stating what governance territory the aspect covers; a collection of member cells; and integration into a Self through the three-tier hierarchy. An aspect is a governance structure within a Self — the middle tier between individual cells and the integrated Self. Multiple aspects can coexist within one Self, and cells can be members of more than one aspect. The aspect is an intra-Self organizational unit: it describes how cells are grouped and coordinated within a single Self under that Self’s governance.

Paper 3 definition.

At Paper 3 scope, the aspect retains its full Paper 2 definition as an intra-Self coordination unit. Paper 3 adds an inter-Self role: the aspect is the unit of exchange in Full Aspect Integration. When a Self contributes content to the shared substrate during an FAI event, it contributes aspects — not individual cells and not the full Self. The contributed aspect surfaces its constituent cells’ DNA-layer and action-layer content within the shared substrate. LLM weights and instinct-layer content do not exchange; the exchange is bounded to substrate content per the instinct/reasoning separation. The aspect’s two roles at Paper 3 scope — intra-Self coordination unit and inter-Self exchange unit — coexist. Context determines which role is active: an aspect within the sharing scope of an active FAI event is operating as an exchange unit; an aspect outside that scope continues to operate as a Paper 2 intra-Self coordination unit.

What changes and what does not.

The aspect’s definition — a governed coordination unit grouping cells around a shared content-domain, with its own DNA and action layers — does not change between Paper 2 and Paper 3. What changes is the scope at which the aspect operates: Paper 2 establishes the intra-Self role; Paper 3 adds the inter-Self role. The exchange unit role is additive. Paper 3 does not replace the Paper 2 definition; it extends the architectural territory in which the same object operates.

4. Contributed Aspect: Paper 3 Context-Specific Usage

What “contributed aspect” names.

A contributed aspect is an aspect that has been contributed to a shared substrate during an FAI event. The “contributed” qualifier marks a specific condition — active participation in a shared substrate — not a new architectural tier. A contributed aspect is a Paper 2 aspect operating at inter-Self scope. The following properties are specific to an aspect in contributed status:

- A contribution record is present in the shared substrate, establishing the provenance of the contribution.
- The aspect’s DNA layer and action layer are present within the shared substrate, accessible under joint governance authority.
- The constituent cells’ content is surfaced through the contributed aspect within the shared substrate (bounded to substrate content; instinct-layer content does not cross the inter-Self perimeter).
- The contributed aspect is subject to the shared substrate’s governance: the joint authority holding inspect, modify, and override rights over shared substrate content holds those rights over the contributed aspect’s content for the duration of the FAI event.

Not all aspects are contributed aspects.

At any given time — including during an active FAI event — only the aspects within a specific event’s sharing scope are contributed aspects. Aspects outside the sharing scope remain ordinary intra-Self coordination units within their home perimeter. The governance-configured sharing scope (which aspects a Self contributes, under what conditions) determines which aspects enter contributed status for a given event. After the FAI event dissolves, the contributed aspects’ content may or may not persist in a durable shared substrate record, depending on governance-configured persistence policy; the contributed status itself ends at dissolution.

How to read Paper 3 usage.

When Paper 3 uses “aspect” without qualification, the intra-Self definition from Paper 2 applies as the baseline. When context establishes that an FAI event is active and an aspect is within the event’s sharing scope, the exchange unit role is also active. “Contributed aspect” is the explicit form for the latter case — the qualifier signals that the aspect has crossed the inter-Self perimeter and is now operating under joint authority within a shared substrate. Practitioners reading Paper 3 should apply this reading rule: *aspect* (unqualified) = Paper 2 intra-Self coordination unit; *contributed aspect* = that same unit in active shared-substrate participation.

5. The Disambiguation: A Compact Summary

Term	Paper 1	Paper 2	Paper 3
<i>Cell</i>	Atomic governance unit; the sole named structural unit	Same definition; DNA/action layers named; placed at base of three-tier hierarchy	Same definition; participates in FAI indirectly through parent aspect
<i>Aspect</i>	Not present	Intra-Self coordination unit grouping cells around a content-domain	Same intra-Self definition PLUS exchange unit role in FAI events
<i>Contributed aspect</i>	Not present	Not present	Paper 3 context-specific: an aspect in active shared-substrate participation under joint authority

Three architectural commitments follow from this table:

- **Cell consistency.** The cell’s governing definition is Paper 1’s. Papers 2 and 3 build on it. No Paper 2 or Paper 3 usage of “cell” introduces a new architectural object; every cell in the trilogy is the Paper 1 atomic governance unit.
- **Aspect additive extension.** The aspect’s governing definition is Paper 2’s. Paper 3 adds the exchange unit role without replacing or narrowing the Paper 2 definition. The two roles coexist; context determines which is active.
- **Contributed aspect as qualifier, not tier.** “Contributed aspect” is not a third tier in the hierarchy between aspect and Self, nor a new kind of aspect distinct from Paper 2’s aspect. It is a qualifier for the specific condition of an aspect in active FAI participation. Outside that condition, the object is a Paper 2 aspect, and should be read as one.

6. Prior-Art Disambiguation Claim

This note establishes the following prior-art positions:

- **On cell.** The cell as the atomic governance unit of a CKS architecture — with a governance substrate, orchestration rules, AI-as-mediator, and human governance authority holding inspect/modify/override rights — is established by Paper 1. Papers 2 and 3 do not introduce a new cell concept. Any implementation of a CKS-pattern cell at any scope (Paper 1 single-cell, Paper 2 cell-within-a-Self, Paper 3 cell-within-an-FAI-participating-Self) is an implementation of the Paper 1 cell. Prior art for the cell architecture extends from Paper 1 through the full trilogy without redefinition.

- **On aspect.** The aspect as a governed coordination unit grouping cells around a shared content-domain — with its own DNA layer and action layer — is established by Paper 2. Paper 3 extends the aspect to the inter-Self exchange role in FAI without replacing the Paper 2 definition. Prior art for the aspect as intra-Self coordination unit is Paper 2. Prior art for the aspect as the unit of inter-Self content exchange in a shared-substrate FAI mechanism is Paper 3, with the intra-Self definition carrying through.
- **On contributed aspect.** The contributed aspect — an aspect in active participation in a shared substrate under joint authority — is a Paper 3 context-specific usage of the Paper 2 aspect concept. Prior art for this usage is Paper 3. “Contributed aspect” does not name a new architectural tier and should not be read as introducing one.
- **What this forecloses.** An adversary cannot claim novelty in: (a) a cell architecture identical to or derived from Paper 1’s atomic governance unit; (b) a coordination unit that groups cells around a content-domain with governed DNA and action layers, as described in Paper 2; (c) using that coordination unit as the exchange granularity in an inter-Self or inter-deployment content exchange mechanism, as described in Paper 3; or (d) a “contributed” or “shared” form of such a coordination unit that operates under joint governance authority in a temporary shared medium. All four of these positions are covered by the trilogy as prior art, with Paper 1 as the earliest priority date for cell-level commitments and Paper 2 as the earliest priority date for aspect-level commitments.

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